

Assessment Tool - 1

SHORT ANSWERS

COURSE CODE	CSA0702	COURSE TITLE	COMPUTER NETWORKS
CO - ASSESSED	CO1	ASSESSMENT	AT-1 : SHORT ANSWERS
WEIGHTAGE		TOTAL MARKS	20

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SECTION DATE	CSE 4 2025	DATE OF SUBMISSION	17/07/2026
FACULTY IN CHARGE	Dr. R. S. Selvan <i>[Signature]</i>	ASSESSMENT	Individual Submission

TOTAL MARKS	20	OBTAINED MARKS	20
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Course
Coordinator

HOD

understanding of concept	Explanation	Application	organisation	accuracy	clarity
25%	25%	20%	15%	10%	5%
5	5	4	3	2	1

1) File Transfer using Transport & Data Link Layer

Given:

$$\begin{aligned}\text{Full size} &= 150 \text{ MB} = 150 \times 10^6 \\ &= 150000000 \text{ bytes}\end{aligned}$$

$$\text{segment size} = 1500 \text{ bytes}$$

$$\text{Frame overhead} = 40 \text{ bytes}$$

$$\text{Bandwidth} = 50 \text{ mbps}$$

calculation:

$$\begin{aligned}\text{No. of segments} &= 150,000,000 / 1500 \\ &= 100000 \text{ segments}\end{aligned}$$

$$\text{No. of frames} = 100000 \text{ frames}$$

$$\begin{aligned}\text{Total overhead} &= 100,000 \times 40 \\ &= 4000000 \text{ bytes (4 MB)}\end{aligned}$$

$$\text{Total transmitted data} = 150 + 4 = 154 \text{ MB}$$

$$\begin{aligned}\text{Transmission Time} &= (154 \times 8) / 50 \\ &= 24.64 \text{ seconds.}\end{aligned}$$

2) Sliding Window Protocol

Given:

Window size = 6 frames

Total frames = 48

Calculations:

$$\begin{aligned}\text{Window required} &= 48 / 6 \\ &= 8 \text{ windows}\end{aligned}$$

One frame lost per window = 8 retransmission

Explanation:

Flow control prevents the sender from overwhelming the receiver, avoids buffer overflow, improves reliability, and increases communication efficiency.

3) IP Fragmentation

Given:

packet size = 12000 bytes

$$MTU = 1500 \text{ bytes}$$

$$IP \text{ header} = 20 \text{ bytes}$$

calculation :

$$\begin{aligned} \text{Data per Fragment} &= 1500 - 20 \\ &= 1480 \text{ bytes} \end{aligned}$$

$$\begin{aligned} \text{Fragments} &= 12000 / 1480 \\ &= 9 \end{aligned}$$

$$\begin{aligned} \text{Header overhead} &= 9 \times 20 \\ &= 180 \text{ bytes} \end{aligned}$$

Explanation :

The network layer fragments packet to fit the MTU. Each fragment carries its own header, and the destination reassembles them using identification offset, and more - fragment fields.

4) Sliding Window Protocol

Given :

$$\text{Sliding Window size} = 6 \text{ frames}$$

Each frame carries = 1024 bytes of user data

Total frame transmitted = 48 frames

one frame in every window is lost

calculation :

$$\text{NO. OF WINDOWS} = 48 \div 6 = 8$$

$$\text{Total retransmissions} = 8 \text{ frame}$$

Explanation :

Flow control regulates the transmission rate, reduces congestion, prevents packet loss, and improves overall network performance.

5) Session Layer Synchronization

Given :

File size = 600 MB

Checkpoint every 60 MB

Failure after 390 MB

calculation

$$\begin{aligned}\text{checkpoints before failure} &= 390 / 60 \\ &= 6 \text{ checkpoints}\end{aligned}$$

$$\text{Last checkpoint} = 360 \text{ MB}$$

$$\begin{aligned}\text{Data to retransmit} &= 390 - 360 \\ &= 30 \text{ MB}\end{aligned}$$

Explanation:

The session layer inserts synchronization points so that after a failure, transmission resumes from last checkpoint instead restarting from beginning.

6) Presentation Layer:

Given:

$$\text{original file} = 900 \text{ MB}$$

$$\text{compression} = 40\%$$

$$\text{Encryption overhead} = 5\%$$

calculation:

$$\begin{aligned}\text{compressed size} &= 900 \times 60\% \\ &= 540 \text{ MB}\end{aligned}$$

$$\begin{aligned}\text{Encrypted size} &= 900 \times 60\% \\ &= 540 \text{ MB}\end{aligned}$$

$$\begin{aligned}\text{Total reduction} &= \left(\frac{900 - 540}{900} \right) \times 100 \\ &= 37\%\end{aligned}$$

Explanation:

The presentation layer compress data to reduce transmission time and encrypts it to ensure confidentiality and secure communication.

7) FTP Data Transfer

Given:

$$\text{Data} = 3 \text{ GB} = 3000 \text{ MB}$$

$$\text{Request size} = 3 \text{ MB}$$

$$\text{Throughput} = 120 \text{ mbps}$$

calculation:

$$\text{Requests} = 3000 / 3 = 1000 \text{ requests}$$

$$\begin{aligned}\text{Data in megabits} &= 3000 \times 8 \\ &= 24000 \text{ Mb}\end{aligned}$$

$$\begin{aligned}\text{Transmission time} &= 24000 / 120 \\ &= 200 \text{ seconds}\end{aligned}$$

Explanation:

The Application Layer provides services such as FTP, manages files transfer, authentication, and ensures reliable communication between users and applications.

8) End To End Delay

Given:

$$\text{Application} = 4 \text{ ms}$$

$$\text{Transport} = 3 \text{ ms}$$

$$\text{Network} = 5 \text{ ms}$$

$$\text{Data link} = 2 \text{ ms}$$

$$\text{Physical} = 1 \text{ ms}$$

$$\text{Propagation} = 18 \text{ ms}$$

$$\text{Transmission} = 12 \text{ ms}$$

calculation :

$$\begin{aligned}\text{Total delay} &= 4 + 3 + 5 + 2 + 1 + 18 + 12 \\ &= 45 \text{ ms}\end{aligned}$$

Explanation :

Each OSI Layer performs a specific function, while propagation and transmission delays contribute to the total end-to-end communication delay.

9) Frame overhead and Efficiency

Given :

$$\text{Useful data} = 80 \text{ MB} = 80000000 \text{ bytes}$$

$$\text{Frame size} = 1600 \text{ bytes}$$

$$\text{Overhead} = 80 \text{ bytes}$$

calculation :

$$\begin{aligned}\text{Useful data per frame} &= 1600 - 80 \\ &= 1520 \text{ bytes}\end{aligned}$$

$$\begin{aligned}\text{Frames required} &= 80000000 / 1520 \\ &\approx 52632 \text{ frames}\end{aligned}$$

$$\text{Total Overhead} = 52632 \times 80 = 4210560 \text{ bytes}$$

$$\approx 4.21 \text{ MB}$$

$$\begin{aligned} \text{Transmission efficiency} &= (1520/1600) \times 100 \\ &= 95\% \end{aligned}$$

Explanation:

Protocol overhead reduces the effective data rate because additional bytes are transmitted for headers and trailers, lowering network efficiency.

10) Hospital medical Record Transfer

Given:

$$\text{Original file} = 240 \text{ MB}$$

$$\text{Compression} = 30\%$$

$$\text{segment size} = 1400 \text{ bytes}$$

$$\text{Frame overhead} = 60 \text{ bytes.}$$

calculation :

$$\text{compressed size} = 240 \times 70\%$$

$$= 168 \text{ MB}$$

$$\text{compressed bytes} = 168,000,000 \text{ bytes}$$

$$\text{segments} = 168,000,000 / 1400$$

$$= 120,000 \text{ bytes } (7.2 \text{ MB})$$

$$\text{Protocol overhead} = 120,000 \times 60$$

$$= 7,200,000 \text{ bytes } (7.2 \text{ MB})$$

$$\text{Final Transmitted data} = 168 + 7.2$$

$$= 175.2 \text{ MB}$$

Explanation :

The presentation Layer compresses data. The Transport Layer divides it into segments. The Data link layer adds frame overhead and the physical layer transmits the data securely and reliably.

Q. 1.1 MB